

1 Related Work

Hong and Oh [Hong2018] provided bounds on the number of polygon mosaics.

Theorem 1.1 ([Hong2018]). *The number of (m, n) polygon mosaics for $m, n \geq 3$ is between $2^{m+n-3} \left(\frac{17}{10}\right)^{(m-2)(n-2)}$ and $2^{m+n-3} \left(\frac{31}{16}\right)^{(m-2)(n-2)}$.*

Our Theorem ?? compliments this result with a procedure to exactly count polygon mosaics, as well as recreate the sequence A181245 in the OEIS [oeis]. Note that unlike here, Hong and Oh did not consider the mosaic made from only blank tiles a polygon mosaic, and so our result differs from theirs by 1.

The methods in [Oh2014] could be used to count polygon mosaics, a fact which is likely known to the authors of [Oh2014], though to our knowledge has not appeared in print. However, their methods are incapable of enumerating our newly introduced polygon free mosaics. Our work extends their methods so we can count both polygon mosaics and polygon free mosaics.

[[Let's be consistent about using the past tense.]] In related work, Lomonaco and Kauffman [Lomonaco08] introduced mosaics constructed from the eleven distinct tile types of Figure ??, the first seven of which form our tile set. [[Question: did Lomonaco and Kauffman also provide bounds?]] They focused on *knot mosaics*, which are mosaics that can be made using these eleven tile types in which every end of a dotted line segment on a tile meets, on an adjacent tile, another end of a dotted line segment. This is the same criterion which we used in Section (??) to characterize polygon mosaics. Oh et al. [Oh2014] used an analogous method to that of our Theorem ?? to count the number of knot mosaics, a topic we will discuss further in the next section. Oh and colleagues went beyond counting by also bounding the growth rate of knot mosaics [Choi2024, Oh2016, Oh2019], and Oh further adapted techniques similar to Theorem ?? to solve problems in monomer and dimer tilings [Oh2018Aztec, Oh2019tiling]. Related ideas were independently used to count the number of rectangular partitions of a rectangle in [oeispaper] for the sequence A182275 in the OEIS [oeis].

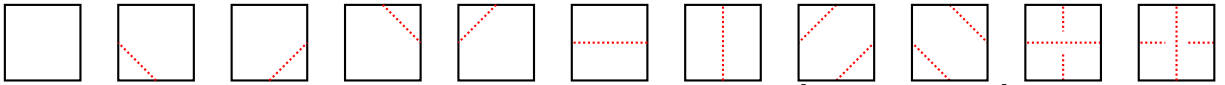


Figure 1: The Lomonaco and Kauffman tile set [Lomonaco08]

2 Extensions and an Open Question

It is simple to adapt our techniques to count polygon mosaics and polygon free mosaics in certain extensions of our tile set. Throughout this section, fix positive integers m and n , and let $\sum_{\ell} v(\ell)$ always designate the sum of $v(\ell)$ over all (m, n) framed lattices ℓ . In all cases, v is a cell product function, and therefore Theorem ?? tells us that matrices can be used to evaluate these sums.

Fewer tiles. If we limit ourselves to a subset of the tiles in Figure ?? then, of course, there are fewer polygon mosaics and polygon free mosaics. We can count how many by modifying our existing work. For example, omitting the rightmost tile of Figure ??, the number of polygon mosaics equals $\sum_{\ell} v(\ell)$, where v is the cell product function that maps $\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}$, $\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}$, $\begin{smallmatrix} 0 & 1 \\ 0 & 1 \end{smallmatrix}$, and $\begin{smallmatrix} 1 & 0 \\ 1 & 0 \end{smallmatrix}$ to 0 and the remaining twelve cells to 1. This can be shown using the reasoning of Theorem ??, since the omitted tile is the one that maps via $\mathcal{L}$ to either $\begin{smallmatrix} 0 & 1 \\ 0 & 1 \end{smallmatrix}$ or $\begin{smallmatrix} 1 & 0 \\ 1 & 0 \end{smallmatrix}$.

We can similarly count the number of polygon mosaics that can be formed by omitting a different tile or by omitting several tiles. It is worth noting, however, that some of these turn out to be trivial. Specifically, if the second, third, fourth, or fifth tile from left is omitted from Figure ?? then there is only one possible polygon mosaic, namely the one that contains only blank tiles. We outline three ways to see this when omitting the second tile; omitting the third, fourth, or fifth tile from left is similar. First assume a polygon mosaic has tiles that are not blank. Of these tiles, look at those in the highest possible row. The one farthest to the right must be the second tile of Figure ?. Second, in any framed binary lattice in which not all vertices are zero, among the vertices that are 1 and are as high as possible, the one farthest to the right must be the corner of a cell that is $\begin{smallmatrix} 0 & 0 \\ 1 & 0 \end{smallmatrix}$. Finally, leaving the details to the reader, in Theorem ??, if $v(\begin{smallmatrix} 0 & 0 \\ 1 & 0 \end{smallmatrix}) = 0$ and $v(\begin{smallmatrix} 0 & 0 \\ 0 & 0 \end{smallmatrix}) = 1$ then for all positive integers m and n , the upper right entry of A_n^m equals 1.

As for determining the number of polygon free (m, n) mosaics using a reduced tile set, this will simply be 7^{mn} if the second, third, fourth, or fifth tile is omitted from the tile set, since we have just seen that no polygons can be formed. For these or other omitted tiles, we can use the ideas from Theorem ?. For example, if we omit the last tile of Figure ?? then we use the cell product function v that maps $\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}$, $\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}$, $\begin{smallmatrix} 0 & 1 \\ 0 & 1 \end{smallmatrix}$, and $\begin{smallmatrix} 1 & 0 \\ 1 & 0 \end{smallmatrix}$ to 0, $\begin{smallmatrix} 0 & 0 \\ 0 & 0 \end{smallmatrix}$ and $\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}$ to 6 (since there are six tiles that can replace the blank tile), $\begin{smallmatrix} 0 & 0 \\ 1 & 0 \end{smallmatrix}$ and $\begin{smallmatrix} 1 & 0 \\ 1 & 1 \end{smallmatrix}$ to -1 , and the remaining eight cells to 1. For omitting the blank tile instead, we would use v defined by mapping $\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}$ and $\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}$ to 0, $\begin{smallmatrix} 0 & 0 \\ 0 & 0 \end{smallmatrix}$ and $\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}$ to 6, $\begin{smallmatrix} 0 & 0 \\ 1 & 0 \end{smallmatrix}$ and $\begin{smallmatrix} 1 & 0 \\ 1 & 1 \end{smallmatrix}$ to -1 , and the remaining ten tiles to 1. In this case we would consider polygon mosaics that might include blank tiles, and create only those polygon free mosaics in which the blank tile was replaced by a tile that is not blank. This problem

was posed by the first author of this paper and served as the motivation for all our work. The number of polygon free mosaics with the blank tile omitted is A364440 in the OEIS [oeis].

Adding tile variations. Adding tiles with the same connection points is similar. For instance, consider including three types of tiles that connect the top and bottom midpoints such as in Figure ??.

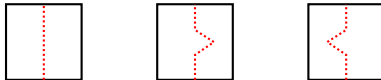


Figure 2: Various tiles that connect the top and bottom midpoints

To count polygon mosaics in this new tile set, take v such that $\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}$ and $\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}$ map to 0, $\begin{smallmatrix} 0 & 1 \\ 0 & 1 \end{smallmatrix}$ and $\begin{smallmatrix} 1 & 0 \\ 1 & 0 \end{smallmatrix}$ map to 3, and all other cells map to 1. To count polygon free mosaics, take v that maps $\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}$ and $\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}$ to 0, $\begin{smallmatrix} 0 & 1 \\ 0 & 1 \end{smallmatrix}$ and $\begin{smallmatrix} 1 & 0 \\ 1 & 0 \end{smallmatrix}$ to 3, $\begin{smallmatrix} 0 & 0 \\ 0 & 0 \end{smallmatrix}$ and $\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}$ to 9, $\begin{smallmatrix} 0 & 0 \\ 1 & 0 \end{smallmatrix}$ and $\begin{smallmatrix} 1 & 0 \\ 1 & 1 \end{smallmatrix}$ to -1 , and map the eight remaining cells to 1.

One could also construct a tile set with both styles of tiles in Figure ?. If this is done for each tile that joins adjacent sides of the unit square, then to count polygon mosaics take v such that $\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}$ and $\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}$ map to 0, $\begin{smallmatrix} 0 & 1 \\ 0 & 1 \end{smallmatrix}$, $\begin{smallmatrix} 1 & 0 \\ 1 & 0 \end{smallmatrix}$, $\begin{smallmatrix} 0 & 0 \\ 1 & 1 \end{smallmatrix}$, $\begin{smallmatrix} 0 & 0 \\ 0 & 0 \end{smallmatrix}$, and $\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}$ map to 1, and all other cells map to 2. To count polygon free mosaics, take v that maps $\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}$ and $\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}$ to 0, $\begin{smallmatrix} 0 & 1 \\ 0 & 1 \end{smallmatrix}$, $\begin{smallmatrix} 1 & 0 \\ 1 & 0 \end{smallmatrix}$, $\begin{smallmatrix} 0 & 0 \\ 1 & 1 \end{smallmatrix}$, and $\begin{smallmatrix} 1 & 1 \\ 0 & 0 \end{smallmatrix}$ map to 1, $\begin{smallmatrix} 0 & 0 \\ 0 & 0 \end{smallmatrix}$ and $\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}$ to 11, $\begin{smallmatrix} 0 & 0 \\ 1 & 0 \end{smallmatrix}$ and $\begin{smallmatrix} 1 & 0 \\ 1 & 1 \end{smallmatrix}$ to -2 , and map the six remaining cells to 2.

The Lomonaco and Kauffman tile set. Let us explore what we can do with the Lomonaco and Kauffman tile set of Figure ?. First we consider using only the first eight tiles, i.e., our usual seven tiles of Figure ? plus the tile that connects the top and left edges and also the bottom and right edges. Here we use “new polygon mosaic” in this more general sense, i.e., mosaics in which each tile is one of these eight type and in which every line segment is part of a polygon. Our $\mathcal{L}$, as already defined, extends to these new polygon mosaics, with the same reasoning showing that $\mathcal{L}$ now gives a bijection, except from the set of (m, n) new polygon mosaics to the set of (m, n) framed lattices, as opposed to (m, n) proper lattices. There are $2^{(m-1)(n-1)}$ (m, n) polygon mosaics, which we can see either by directly counting the number of (m, n) proper lattices or by computing $\sum_{\ell} v(\ell)$ using the cell product function v that maps all cells to 1. From here, we can easily count the number of knot mosaics by simply adding tile variations. We can uniquely get any knot mosaic by taking a new polygon mosaic and, wherever the eighth tile appears, use any of the last four tiles. Therefore, we can count the number of knot mosaics we use the cell product function that maps $\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}$ and $\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}$ to 4, and maps the fourteen remaining cells to 1. Calculating in this way gives exactly the same number of knot mosaics

as calculating using Oh and colleagues’ Theorem 1, their main result. We should mention, however, that their calculated results, presented just after their statement of Theorem 1, are not quite right. The $n = 7$ case is off by 2 and each later line is off by more, though always by a small percent. We speculate that the errors are due to the authors using 64 bit fixed-precision arithmetic rather than arbitrary-precision arithmetic.

We leave the readers with an open question, which is to count the number of “knot free mosaics,” i.e., mosaics using the Lomonaco and Kauffman tile set of Figure ?? that contain no knots. The techniques we used to count polygon free mosaics do not seem to generalize to this question. One problem is that when we interchange the eighth Lomonaco and Kauffman tile for one of the remaining tiles, the number of knots can change. This suggests that there may be no way to count the number of knots based on local information in the manner of Proposition ??.

3 Acknowledgment

The authors would like to thank Richard Schank for contributing ideas toward the definition of $\mathcal{L}$ and ideas that led to Theorem ??.

4 Appendix

Proposition (The inclusion-exclusion principle). *For finite sets $A_1, A_2, \dots, A_M$,*

$$\left| \bigcup_{j=1}^M A_j \right| = \sum_{\emptyset \neq I \subseteq \{1, 2, \dots, M\}} (-1)^{|I|-1} \left| \bigcap_{j \in I} A_j \right|.$$

Proof. The $M = 1$ case is trivial, and $M = 2$ case is the familiar and easily justified statement that $|A_1 \cup A_2| = |A_1| + |A_2| - |A_1 \cap A_2|$. We proceed inductively, so for $M \geq 3$, assume the result for $M - 1$ sets, and let us prove it for M sets. We split up the sum on the right of our desired equation into sets I that do not contain M and those that do. By our inductive hypothesis, the first sum equals $\left| \bigcup_{j=1}^{M-1} A_j \right|$. The second sum is $\sum_{\{M\} \subseteq I \subseteq \{1, 2, \dots, M\}} (-1)^{|I|-1} \left| \bigcap_{j \in I} A_j \right|$, which we rewrite by separating the $I = \{M\}$ term and reindexing the rest using $J = I \setminus \{M\}$ to get

$$|A_M| - \sum_{\emptyset \neq J \subseteq \{1, 2, \dots, M-1\}} (-1)^{|J|-1} \left| \bigcap_{j \in J} (A_j \cap A_M) \right|.$$

Again using our inductive hypothesis, this final sum equals $\left| \bigcup_{j=1}^{M-1} (A_j \cap A_M) \right| = \left| \left(\bigcup_{j=1}^{M-1} A_j \right) \cap A_M \right|$. The entire right side of our desired equation therefore equals $\left| \bigcup_{j=1}^{M-1} A_j \right| + |A_M| - \left| \left(\bigcup_{j=1}^{M-1} A_j \right) \cap A_M \right|$. By the $M = 2$ case, this is $\left| \bigcup_{j=1}^M A_j \right|$, as desired. $\square$